

Manual Test Checklist

ECITY — Mobile Shop Management Web App

Version 1.1 · 2026

What a person still has to judge, module by module

Companion documents: PRD, Module Breakdown, Engineering Design, Deployment Guide, Database Guide

How to Use This

Automated tests cover the mechanics — 200 unit and integration tests, 407 browser tests across four screen sizes. This checklist covers what a person still has to judge: does it *feel* right, does the wording make sense, does the shop's actual workflow survive contact with the screen.

Work through a module's section after it is built, and again before go-live. Tick as you go. Anything that fails is a bug, not a note.

One section per module. M1–M14 append to this file as they ship.

Before you start

```
docker compose up -d      # or open Postgres.app
npm run db:migrate
npm run db:seed           # resets the three test accounts
npm run dev               # http://localhost:3000
```

To return to a known-clean state at any point, re-run `npm run db:seed`. It upserts, so it is safe to repeat.

Test accounts

Email	Password	Role	Branch scope
admin@ecity.local	ChangeMe!2026	Admin / Owner	All branches (14 permissions)
manager@ecity.local	ChangeMe!2026	Branch Manager	MAIN only (7 permissions)
staff@ecity.local	ChangeMe!2026	Staff	MAIN only (3 permissions)

Use a **private/incognito window** to hold a second session, so you can be two users at once without signing out.

These accounts must never exist in production. They are flagged `mustChangePassword` and are created only by the seed.

M0 — Foundations

Delivers. PRD FR-1.1 – FR-1.5, FR-31.1, plus branch context and the authorisation layer everything else depends on.

Signing in as the admin exercises perhaps a tenth of this module. The interesting part is what the *other two* users cannot do.

1. Authentication

Start signed out, at `/login`.

- [] Wrong password for a real account → *"Email or password is incorrect."*
- [] A made-up email with any password → **the identical message**. It must not reveal whether the account exists
- [] Empty email or password → inline validation, no request sent
- [] Visit `/settings/users` directly while signed out → redirected to `/login`
- [] Correct credentials → lands on `/dashboard`
- [] Avatar menu → **Sign out** → back at `/login`

- [] After signing out, the browser Back button does **not** restore the app
- [] Sign in again in a second browser → both sessions work independently

Password reset

Email is not wired up until M14, so the reset link is shown on screen in development.

- [] `/forgot-password` → enter `admin@ecity.local` → confirmation appears
- [] A **yellow development box** shows the reset link
- [] Enter an address that does not exist → **the same confirmation**, no hint that the account is unknown
- [] Follow the link, set a new password → redirected to `/login`
- [] The **old** password no longer works
- [] The **new** password works
- [] Open the same reset link a second time → refused as already used
- [] Reset the password back to `ChangeMe!2026`, or re-run `npm run db:seed`

Lockout

- [] Get the password wrong 8 times in a row → *"Too many failed attempts."*
- [] Wait 15 minutes, or `npm run db:seed`, to clear it

2. Branch scoping

The heart of M0. Sign in as each user and open the branch switcher in the header.

- [] **admin** — sees *All branches (consolidated)*, MAIN and NORTH
- [] **manager** — sees **MAIN only**. No NORTH, and no "All branches"
- [] **staff** — sees MAIN only
- [] As admin, switch between branches → the dashboard's *Branch context* card updates (active branch, accessible count, consolidated yes/no)
- [] The chosen branch survives a page reload
- [] On a phone-width window the switcher shows the branch **code**, not the full name, and the header still fits

3. Permissions

Check what is **missing**, not what is present.

User	Sidebar should show
admin	Dashboard, Users, Roles, Audit log
manager	Dashboard, Users, Audit log — no Roles
staff	Dashboard only

- [] Each user's sidebar matches the table
- [] As **staff**, type `/settings/roles` into the address bar → redirected, not shown
- [] As **staff**, type `/settings/users` → redirected

- [] As **manager**, `/settings/roles` → redirected
- [] The dashboard's *Your permissions* card lists 14 / 7 / 3 entries respectively

4. Server-side authorisation

A hidden menu item is not a control. This step proves the server refuses the request, not just that the button was hidden.

Sign in as **staff** in the browser, then copy the session cookie from DevTools → Application → Cookies → `ecity_session`, and run:

```
COOKIE='ecity_session=PASTE_VALUE_HERE'
curl -s -o /dev/null -w '%{http_code}\n' localhost:3000/api/users -H "Cookie: $COOKIE"
curl -s -o /dev/null -w '%{http_code}\n' localhost:3000/api/audit -H "Cookie: $COOKIE"
curl -s -o /dev/null -w '%{http_code}\n' localhost:3000/api/roles -H "Cookie: $COOKIE"
```

- [] All three return **403**, not data
- [] The body reads `{"error":"...","code":"FORBIDDEN"}`
- [] With no cookie at all → **401**
- [] As **manager**, forcing another branch is refused:

```
curl -s -o /dev/null -w '%{http_code}\n' -X POST localhost:3000/api/session/branch \
-H "Cookie: $COOKIE" -H 'content-type: application/json' -d '{"branchId":2}'
```

- [] Returns **403** — branch NORTH is not theirs
- [] `{"branchId":null}` (the consolidated view) also returns **403**

5. User management

As **admin**, Users → Add user.

- [] Password under 10 characters → rejected inline
- [] An email that already exists → *"A user with this email already exists."*
- [] No role chosen → *"Choose a role."*
- [] Create a Staff user assigned to **NORTH** only
- [] They appear in the user list with the right role and branch badge
- [] Sign in as the new user → branch switcher shows **NORTH only**
- [] Their first sign-in is flagged "must change password"

Roles

As **admin**, Settings → Roles.

- [] Select **Staff**, tick *View the audit log*, Save
- [] Sign in as staff → **Audit log now appears** in their sidebar
- [] Any staff session open elsewhere was **signed out** when the role changed, rather than keeping stale permissions until next login
- [] Untick it and Save again
- [] Try to remove *Create and edit users* from **Admin** → refused, so the business cannot lock itself out
- [] System roles cannot be deleted

6. Audit log

As **admin**, Settings → Audit log.

- [] Everything done above appears: **LOGIN**, **LOGIN_FAILED**, **LOGOUT**, **CREATE**, **PERMISSION_CHANGED**, **PASSWORD_RESET_COMPLETED**
- [] Each row shows who, what, when
- [] Expanding a "*n field(s)*" row shows old → new values
- [] A password hash **never** appears in a changes payload
- [] Paging works if there are more than 50 entries

The log is append-only, enforced by the database rather than by code:

```
psql ecity -c "update audit_log set summary='tampered' where id=1;"
psql ecity -c "delete from audit_log where id=1;"
```

- [] Both fail with `audit_log is append-only`

7. Responsive layout

Resize the window, or use DevTools device mode. Check at ~375px, ~810px and full width.

- [] Below ~768px the sidebar is replaced by a **hamburger** top-left
- [] Tapping it opens a drawer; choosing a destination closes it
- [] On **Users**, the table becomes **one card per user**
- [] On **Audit log**, entries become cards
- [] Nothing scrolls sideways at any width
- [] Buttons and links are comfortably tappable with a thumb
- [] The **Roles** editor is usable — role list becomes a horizontal strip
- [] Forms stack, with full-width buttons

8. Theme

- [] Primary colour is **black** throughout (buttons, active nav item)
- [] Set `APP_THEME=navy` in `.env`, restart → the whole app is blue, with no other visual change
- [] An invalid value such as `APP_THEME=nonsense` → falls back to black rather than breaking
- [] Set it back to `onyx`

What M0 deliberately does not include

Do not raise these as bugs:

- No products, inventory, IMEIs, customers or suppliers — those are M1 and M2
- The dashboard shows branch context and permissions, **not** sales figures. The real dashboard is M10
- No branch create/edit screen — M1. The two seeded branches are fixed
- Password reset emails print to screen instead of sending — M14

- Notifications, reports, exports — later modules

M1 — Master Data

Delivers. PRD FR-2.1 – FR-2.4, FR-3.1, FR-3.2, FR-3.8, FR-5.10, FR-6.6, FR-10.1.

The shop gets configured and the people it trades with exist. Nothing here moves stock or money yet — these are the pickers and directories the rest of the system draws on.

Customer = someone who buys from the shop. **Supplier** = someone the shop buys from. Both are shared across every branch.

1. Business settings

Sign in as **admin**, go to **Settings** → **Business**.

- ☐ Four tabs: Profile, Tax, Payments, Expenses
- ☐ On a phone the tabs scroll sideways rather than wrapping

Profile

- ☐ Edit the business name, save, reload — the change persisted
- ☐ Enter a malformed GST number → *"Enter a valid 15-character GSTIN."*
- ☐ Enter a valid one (29ABCDE1234F1Z5) → accepted, stored upper-case
- ☐ Currency is fixed at INR and cannot be edited
- ☐ Toggle **Prices include tax** → the explanation below it changes
- ☐ Save, reload → the toggle held its new position

Tax

- ☐ Five GST slabs are listed: 0, 5, 12, 18, 28%
- ☐ **GST 18%** is marked Default
- ☐ Rates display as 18.00%
- ☐ Add a rate named GST 3% at 3 → appears in the list
- ☐ Add one with the same name → refused as a duplicate
- ☐ Deactivate a non-default rate → shows Inactive
- ☐ Try to deactivate the **default** rate → refused, with the reason

Payments

- ☐ Cash, UPI, Card, Bank Transfer are listed with their type badge
- ☐ Cash carries a **Cash drawer** badge; the others do not
- ☐ Deactivate UPI → shows Inactive; reactivate it
- ☐ Try to deactivate **Cash** → refused: at least one cash method must stay active

Expenses

- ☐ Seven categories: Rent, Electricity, Salaries, Transport, Packaging, Repairs, Miscellaneous
- ☐ Add one → appears immediately

- [] Add the same name again → refused

2. Branches

Settings → **Branches** as admin.

- [] MAIN and NORTH are listed with manager and user counts
- [] **Add branch** — code **MGROAD**, name **MG Road** → created and listed
- [] The code is forced to upper case even if typed lower
- [] Add another with code **MGROAD** → *"Branch code MGROAD is already in use."*
- [] A code containing a space or ! is rejected inline
- [] Open the new branch, change its name, save → the list shows the new name
- [] Assign a manager from the dropdown → shows in the Manager column
- [] Leave the manager empty → **saves without error** (*this failed silently before — see §5*)

Deactivation — the rule that matters (FR-3.8)

- [] Deactivate **MG Road** → badge turns Inactive, **the row stays**
- [] Open the header **branch switcher** → MG Road is **gone**
- [] Reactivate it → it comes back in the switcher
- [] Deactivate every branch but one, then try the last → refused: at least one branch must stay active

3. Customers and suppliers

Customers in the sidebar.

- [] The list is empty, or holds only what you have created
- [] **Add customer** with a name only → saves. A walk-in has nothing else
- [] Add one with phone **9876543210**
- [] Add a second with the **same phone** → *"Another customer (...) already uses this phone number."*
- [] Give a **supplier** that same phone → **allowed**, they are different directories
- [] Edit a customer, keep their phone unchanged, save → no false duplicate
- [] Enter **not-a-phone** → rejected inline
- [] Enter **MiXeDqExample.COM** → stored lower-case (check in Studio)

Search and status

- [] Search by part of a name → matches
- [] Search by phone → matches
- [] Search by email → matches
- [] Search by GST number → matches
- [] **Deactivate** a customer → disappears from the default list
- [] Click **Show inactive** → they reappear, badged Inactive
- [] Reactivate → back to normal

Repeat the same pass on **Suppliers**. Suppliers additionally have a **Company** field; customers do not.

4. Permissions

User	Should see in the sidebar
admin	Customers, Suppliers, Business, Branches, Users, Roles, Audit log
manager	Customers, Suppliers, Business, Users, Audit log — no Branches, no Roles
staff	Customers, Suppliers — no Settings group at all

- [] Each user's sidebar matches
- [] As **staff**, open `/settings/business` directly → redirected
- [] As **staff**, open `/settings/branches` directly → redirected
- [] As **staff**, open `/customers/new` directly → redirected (they may view customers, not create them)
- [] As **manager**, `/settings/branches` → redirected. Managers run a branch; they do not create them

At the API, signed in as staff (see M0 §4 for how to get the cookie):

```
curl -s -o /dev/null -w '%{http_code}\n' localhost:3000/api/branches?manage=1 \
-H "Cookie: $COOKIE"
curl -s -o /dev/null -w '%{http_code}\n' -X POST localhost:3000/api/customers \
-H "Cookie: $COOKIE" -H 'content-type: application/json' -d '{"name":"Sneaky"}'
```

- [] The POST returns **403**

5. Form errors are never silent

A bug found during M1: a validation failure with no visible message meant the Save button appeared to do nothing.

- [] Paste 300 characters into any address field → a visible error appears
- [] On every form, an invalid field shows red text under **that** field
- [] No form ever fails with the button simply doing nothing

6. Audit

Settings → **Audit log** as admin.

- [] Everything above is recorded: branch created, customer created, tax rate added, payment method deactivated
- [] Entity types include `branch`, `customer`, `supplier`, `tax_rate`, `payment_method`, `business`
- [] Expanding a row shows old → new values

7. Responsive

At ~375px, ~810px and full width:

- [] Customers, Suppliers and Branches lists become **cards** below 768px
- [] Business settings tabs scroll rather than wrap
- [] Forms stack to one column; buttons go full width

- [] Nothing scrolls sideways
- [] The sidebar's **Master data** and **Settings** groups appear in the mobile drawer

What M1 deliberately does not include

- **No customer/supplier history tab yet.** Clicking a customer opens the edit form. The profile view showing their purchases, returns and credit arrives with M4/M5, when there is something to show. *This is a known gap against M1's stated criteria, carried into M2.*
 - No products, stock or IMEI's — that is M2
 - Attachments have a working API and storage layer, but **no upload button on any screen yet**; the first screen that needs one is M3's purchase form
 - Logo upload is not wired to the business profile
-

M4 — Sales & Billing

Delivers. PRD FR-6.1 – FR-6.8, FR-26.1, FR-26.3, FR-26.4, FR-38.2, FR-38.3.

This is the screen the shop lives on. Everything before it existed so that this one could work: a bill is built, money is taken, stock drops, and an invoice comes out. Test it standing up, at speed, the way the counter will.

Before you start. You need stock. Run through M3 once to bring in at least one USED or ER handset with an IMEI, and one accessory with a quantity of ten.

1. Speed at the counter

Go to **Billing**.

- [] The search box already has focus — you can start typing without clicking
- [] Type an accessory name → results appear as you type
- [] With exactly one result, pressing **Enter** puts it on the bill
- [] Scan or paste an IMEI → that exact handset lands on the bill
- [] **Time yourself:** accessory → Cash → Save, keyboard only, under 20 seconds
- [] The whole flow needs no mouse

2. The bill

- [] A device line shows the IMEI or serial underneath the product name
- [] A device line shows its main type (USED / ER / ACT / GLOBAL)
- [] A GLOBAL device also shows **NEW CUT** where that applies
- [] A device line's quantity is fixed at 1 and cannot be edited
- [] An accessory line's quantity can be raised, and the total follows
- [] Adding the same IMEI twice is refused with a clear message
- [] Line discount reduces that line only; bill discount reduces the total
- [] **Staff** signed in cannot give a discount at all

- [] Totals: taxable + GST equals the total, exactly, to the paisa
- [] Change a quantity and the totals update instantly

3. The bill survives real life

- [] Build a half-finished bill and **reload the page** → it is still there
- [] Clear bill empties it, and asks nothing you would regret confirming
- [] Switch branch with a bill open → the bill clears (it belonged to the other branch's stock)

4. Payment

- [] **+ Cash** fills the full balance in one press
- [] Split across two methods (say Cash 500 + UPI rest) → both are recorded
- [] Pay less than the total with **no customer attached** → refused, because an unpaid balance needs someone to owe it
- [] Attach a customer, then save partly paid → the sale shows **PARTIAL**
- [] Overpaying is refused

5. Customer, without leaving the bill

- [] Press **New** beside the Customer box
- [] Name and phone only → **Add to bill**
- [] The new customer is created *and* attached, and the bill is untouched
- [] A duplicate phone number is refused with a message naming the clash
- [] Cancel closes the dialog and changes nothing

5c. Pickers do not silently truncate

Customers, suppliers and products are chosen through a **searchable picker**, never a fixed dropdown — a capped list drops records with no hint that it did.

- [] The Customer box on billing opens a search, not a long dropdown
- [] Typing a phone number finds the customer
- [] A customer created minutes ago is findable
- [] Same for **Supplier** on the purchase form
- [] Same for **Product** on the purchase form and on Add device

5b. Customer history (FR-6.7)

Open **Customers** and click a customer who has bought something.

- [] The **History** tab opens first — that is what you came to see
- [] Total spend, number of purchases, outstanding and last purchase date

- ☐ Every invoice is listed, newest first, with branch and item count
- ☐ Clicking an invoice number opens that sale
- ☐ Purchases from **every branch** appear, not just the current one
- ☐ A part-paid bill shows **PARTIAL**, and outstanding is non-zero and red
- ☐ A voided bill is listed but does **not** count towards spend or dues
- ☐ A customer who has bought nothing says so, rather than showing an empty table
- ☐ The **Details** tab still edits the customer and saves
- ☐ On a phone the history becomes cards and nothing overflows sideways

6. The invoice

Save a bill; you land on the sale.

- ☐ The invoice shows shop name, GSTIN, branch, invoice number and date
- ☐ Customer details appear, or "Walk-in customer"
- ☐ Every line shows quantity, price, tax and amount
- ☐ GST is broken out per rate
- ☐ **Print A4** produces a clean full-page invoice with no navigation on it
- ☐ **Print receipt (80 mm)** produces a narrow slip that fits the thermal roll
- ☐ **Download PDF** saves a file named after the invoice number
- ☐ Open the PDF: every figure matches the screen, and no character shows as a blank box
- ☐ The PDF's amounts are labelled INR (the rupee glyph is deliberately not used — see <src/server/pdf/invoice-pdf.tsx>)
- ☐ **Share** hands the PDF to the phone's share sheet (WhatsApp, mail); on a desktop browser it copies the invoice link instead

6b. The invoice is a statutory GST document (OQ-4)

First set **Settings** → **Business** → **GST state**, and a **GST state** on each branch. Give a few products an **HSN code** (8517 phones, 8544 cables).

- ☐ The invoice is headed **Tax Invoice**
- ☐ **Place of supply** appears, with the state code and name
- ☐ Each line shows its **HSN code**
- ☐ An ordinary local sale shows **CGST** and **SGST** as separate lines, each half the rate — never a single combined "GST 18%"
- ☐ CGST + SGST adds up to exactly the tax shown
- ☐ An **HSN summary** table appears below the totals on A4
- ☐ The HSN summary's taxable and tax columns add up to the invoice totals
- ☐ Set a customer's **GST state** to another state, bill them → the invoice shows a single **IGST** line and no CGST/SGST
- ☐ The customer pays the same total either way — only the split changes

- [] A product with no HSN still bills; the line shows "—" and stays out of the summary
- [] The downloaded PDF matches the screen on every one of these

7. Stock actually moved

- [] The handset you sold is now **Sold** on the device page
- [] Its IMEI no longer appears in billing search
- [] The accessory's branch stock dropped by the quantity sold
- [] The stock ledger shows one **SALE** row referencing the invoice

8. The other billing system (FR-38.2, FR-38.3)

- [] A **NEW** handset never appears in billing search at all
- [] Forcing one onto a bill is refused server-side too
- [] Open that device → **Sold in other system** is offered
- [] Mark it → status becomes **SOLD_PENDING_IMPORT** and stock drops now
- [] It disappears from billing search, so it cannot be sold twice
- [] An ECITY-channel device does **not** offer that button

9. Sales list

- [] Every saved sale is listed, newest first
- [] Search by invoice number, customer name or IMEI finds it
- [] Payment status filter: PAID / PARTIAL / UNPAID each return the right rows
- [] Branch filter narrows to that branch; a branch-scoped user sees only theirs
- [] From/To dates include both end days
- [] Filters survive a page reload (they are in the URL, so they are shareable)

10. Concurrency — needs two browsers

- [] Sign in on two windows, put the **same IMEI** on both bills
- [] Save both → the first succeeds, the second fails with a clear message
- [] The device is Sold exactly once, and only one sale exists
- [] Save a bill, then press Save again on a restored copy → still one sale

11. Responsive

- [] On a phone the billing screen is usable one-handed; nothing overflows sideways
- [] The cart and the totals panel stack rather than squeeze
- [] The invoice is readable on a phone

- [] On a large screen the search, cart and payment panel sit side by side

What M4 deliberately does not include

- **No returns or exchanges** — that is M6
 - **No trade-in** on the bill — also M6
 - **No Excel import** of the other system's sales — that is M12; devices marked `SOLD_PENDING_IMPORT` are waiting for it
 - **Returns and payments do not appear in customer history yet** — the tab says so. Returns arrive with M6, payments against dues with M5
 - No daily cash-up or Z-report — M7
-

Cross-cutting — Lists & pagination

Every list in the app is paginated server-side at **25 rows a page**. Check these on **Products, Devices, Customers, Suppliers, Purchases, Sales** and the **Audit log** — the control is the same one everywhere.

- [] Under each list: *"Showing 1–25 of N"*, with the real total
- [] A list with more than 25 rows offers **Previous / Next**
- [] A list with 25 or fewer offers no page buttons, but still states the count
- [] On page 1, **Previous** is greyed out and does nothing when clicked
- [] On the last page, **Next** is greyed out
- [] **Next** advances, and the rows are genuinely different
- [] Page 1's URL has no `?page=` on it — one address for the first page
- [] Apply a filter, then page forward → **the filter is still applied**
- [] Change a filter while on page 3 → you go back to page 1, not an empty page
- [] Copy a page-2 URL into a new tab → same page, same filters
- [] Browser back returns to the previous page of results
- [] On a phone the buttons are big enough to tap and do not overflow sideways

Housekeeping

The end-to-end tests create real branches and customers in your development database. After a test run you will see extra rows. To get back to a clean state:

```
dropdb ecity && createdb ecity
psql -d ecity -c "CREATE ROLE ecity WITH LOGIN PASSWORD 'ecity' SUPERUSER;"
npm run db:migrate && npm run db:seed
```

Sign-off

Tested by	
Date	
Build / commit	
Result	pass / fail
Issues raised	